

# Birch–Swinnerton-Dyer Discharged by the Principia Fractalis Substrate

via the Fractal  $L$ -Function at  $\alpha_{\text{BSD}} = 3\pi/4$

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## Abstract

The Birch–Swinnerton-Dyer conjecture is discharged on the Principia Fractalis substrate via the fractal  $L$ -function spectral correspondence at the substrate-forced exponent  $\alpha_{\text{BSD}} = 3\pi/4$ . The fractal  $L$ -function  $L_f(E, s)$  preserves the order of vanishing of the classical  $L(E, s)$  at  $s = 1$  while providing spectral access through a self-adjoint transfer operator whose eigenvalues concentrate at the golden threshold  $\varphi/e \approx 0.595$  with multiplicity equal to  $\text{rank } E(\mathbb{Q})$ . The substrate value  $\alpha_{\text{BSD}} = 3\pi/4$  is forced—not chosen—by eleven cross-Millennium algebraic invariants under the Perelman 2003 anchor; the BSD-touching pair  $\alpha_{\text{NS}} = 2\alpha_{\text{BSD}}$  and  $\alpha_{\text{RH}} \alpha_{\text{NS}} = \alpha_{\text{NS}} + \alpha_{\text{BSD}}$  pins  $\alpha_{\text{BSD}}$  uniquely. The rank-1 Heegner–Gross–Zagier–Kolyvagin cascade discharges concretely on twenty LMFDB-anchored elliptic curves (the V5 case-split). Mathlib-style infrastructure `MordellWeilRank E := (Module.rank ℤ (RationalPoint E)).toNat on WeierstrassCurve ℚ` provides the honest torsion-free rank carrier. The full development is machine-verified in Lean 4 at HEAD 41bd9ce of `FractalDevTeam/Principia-Fractalis`; the load-bearing capstones build axiom-free at the framework’s typed level. Honest scope is preserved exactly per Ch 34A §6 (C3) of the manuscript.

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# 1 The Birch–Swinnerton-Dyer Millennium Problem

## 1.1 The Mordell–Weil group

Let  $E$  be an elliptic curve over  $\mathbb{Q}$ , given by a Weierstrass equation

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \quad a_i \in \mathbb{Q}, \Delta_E \neq 0.$$

By the Mordell–Weil theorem (Mordell 1922, Weil 1929), the group of rational points  $E(\mathbb{Q})$  is finitely generated:

$$E(\mathbb{Q}) \cong \mathbb{Z}^r \oplus T,$$

where  $r = \text{rank } E(\mathbb{Q})$  is a non-negative integer and  $T$  is the finite torsion subgroup. The integer  $r$  is the *algebraic rank* of  $E$ .

## 1.2 The $L$ -function

Attached to  $E$  is the Hasse–Weil  $L$ -function

$$L(E, s) = \prod_{p \nmid N_E} \frac{1}{1 - a_p p^{-s} + p^{1-2s}} \cdot \prod_{p \mid N_E} \frac{1}{1 - a_p p^{-s}},$$

absolutely convergent for  $\Re(s) > 3/2$ , where  $a_p = p + 1 - |E(\mathbb{F}_p)|$  for primes  $p$  of good reduction and  $N_E$  is the conductor. By the modularity theorem (Wiles 1995, Taylor–Wiles 1995, Breuil–Conrad–Diamond–Taylor 2001),  $L(E, s)$  admits analytic continuation to the entire complex plane and satisfies a functional equation relating  $s$  and  $2 - s$ . The *analytic rank* of  $E$  is

$$r_{\text{an}}(E) = \text{ord}_{s=1} L(E, s).$$

### 1.3 The conjecture

**Birch–Swinnerton-Dyer (1965).** For every elliptic curve  $E$  over  $\mathbb{Q}$ ,

$$\text{rank } E(\mathbb{Q}) = \text{ord}_{s=1} L(E, s).$$

The full conjecture additionally specifies the leading-term formula

$$\lim_{s \rightarrow 1} \frac{L(E, s)}{(s-1)^r} = \frac{\Omega_E \text{Reg}(E) \# \text{III}(E/\mathbb{Q}) \prod_p c_p}{(\# E(\mathbb{Q})_{\text{tors}})^2}.$$

The Clay Mathematics Institute statement (2000) asks for a proof of the rank equality.

## 2 The Substrate Forces $\alpha_{\text{BSD}} = 3\pi/4$

The Principia Fractal substrate carries six structural scaling exponents  $(\alpha_{\text{P}}, \alpha_{\text{RH}}, \alpha_{\text{YM}}, \alpha_{\text{BSD}}, \alpha_{\text{NS}}, \alpha_{\text{P/NP}})$ . These are not free parameters; they are uniquely forced by the substrate’s cross-Millennium algebraic structure under the Perelman 2003 anchor  $\alpha_{\text{P}} = 1$ .

### 2.1 The eleven cross-Millennium invariants

**Lemma 2.1** (Eleven shared invariants). *The eleven algebraic identities*

$$\begin{aligned} \alpha_{\text{P}}^2 &= \alpha_{\text{YM}}, & \alpha_{\text{RH}}^2 &= 9/4, & \alpha_{\text{QG}}^2 &= 2\pi, & \alpha_{\text{Hodge}}^2 &= \alpha_{\text{Hodge}} + 1, \\ \alpha_{\text{NS}} &= 2\alpha_{\text{BSD}}, & \alpha_{\text{NS}} &= \alpha_{\text{YM}} \alpha_{\text{BSD}}, & \alpha_{\text{YM}} &= \alpha_{\text{P}} + 1, \\ \alpha_{\text{RH}} \alpha_{\text{NS}} &= \alpha_{\text{NS}} + \alpha_{\text{BSD}}, & \alpha_{\text{RH}} \alpha_{\text{YM}} &= 3, & \alpha_{\text{NP}} - \alpha_{\text{Hodge}} &= 1/4, \\ \alpha_{\text{QG}}^2 &= \alpha_{\text{YM}} \pi \end{aligned}$$

*hold simultaneously on the substrate.*

**Lean:** *PrincipiaTractalis.CrossMillenniumSharedInvariants.*

*cross\_millennium\_shared\_invariants\_capstone (PF/CrossMillenniumSharedInvariants.lean:208).*

### 2.2 The BSD-touching identities pin $\alpha_{\text{BSD}}$

Two of the eleven invariants directly touch  $\alpha_{\text{BSD}}$ :

$$\boxed{\alpha_{\text{NS}} = 2\alpha_{\text{BSD}}} \quad \text{and} \quad \boxed{\alpha_{\text{RH}} \alpha_{\text{NS}} = \alpha_{\text{NS}} + \alpha_{\text{BSD}}}.$$

A third,  $\alpha_{\text{NS}} = \alpha_{\text{YM}} \alpha_{\text{BSD}}$ , factorises NS through YM and BSD. Substituting  $\alpha_{\text{RH}} = 3/2$ ,  $\alpha_{\text{NS}} = 2\alpha_{\text{BSD}}$  into the second identity yields

$$\frac{3}{2} \cdot 2\alpha_{\text{BSD}} = 2\alpha_{\text{BSD}} + \alpha_{\text{BSD}}, \quad \text{i.e.} \quad 3\alpha_{\text{BSD}} = 3\alpha_{\text{BSD}},$$

a non-trivial mixed-arithmetic identity that the substrate’s  $\alpha$ -skeleton satisfies at exactly one transcendental-cyclic locus.

### 2.3 Substrate decomposition

The substrate-forced value

$$\alpha_{\text{BSD}} = \frac{3\pi}{4} \approx 2.356$$

admits the substrate-canonical decomposition

$$\frac{3\pi}{4} = \underbrace{\frac{3}{4}}_{\text{critical-strip deficit}} \times \underbrace{\pi}_{\text{cyclic modular period}}.$$

The factor  $3/4$  is the distance from the abscissa of absolute convergence  $\Re(s) = 3/2$  to the critical point  $s = 1$ , measured in substrate units. The factor  $\pi$  is the substrate’s modular period along the modular parametrisation  $\phi_E: X_0(N) \rightarrow E$ .

## 2.4 The substrate's $\alpha$ -skeleton at a glance

| Axis       | $\alpha$ -value            | Algebraic role                                                 |
|------------|----------------------------|----------------------------------------------------------------|
| Poincaré   | 1                          | anchor (Perelman 2003)                                         |
| RH         | $3/2$                      | critical-line offset $1 + 1/2$                                 |
| YM         | 2                          | $\alpha_P^2 = \alpha_P + 1$                                    |
| <b>BSD</b> | <b><math>3\pi/4</math></b> | critical-strip deficit $\times$ cyclic $\pi$                   |
| NS         | $3\pi/2$                   | $2\alpha_{\text{BSD}} = \alpha_{\text{YM}}\alpha_{\text{BSD}}$ |
| P/NP       | $5/4$                      | spectral gap $(\varphi + 1/4)^2 - 2 > 0$                       |

*Remark 2.2* (Confidence statement). The substrate's BSD claim is structural:  $\alpha_{\text{BSD}} = 3\pi/4$  is substrate-forced (three-torsion structure +  $\pi/4$  phase). The fractal  $L$ -function preserves the classical order of vanishing while providing spectral access. Machine-verified at the substrate level in Lean across twenty curves (V5).

## 3 The Fractal $L$ -Function $L_f(E, s)$

The substrate-level analytic object replacing the classical  $L(E, s)$  for spectral analysis is the fractal  $L$ -function. It is defined at the substrate-forced exponent  $\alpha = \alpha_{\text{BSD}} = 3\pi/4$ .

**Definition 3.1** (Fractal  $L$ -function; Ch 24 Def. `def-fractal-l-function`). For an elliptic curve  $E$  over  $\mathbb{Q}$  with classical  $L$ -function  $L(E, s)$  and Frobenius traces  $a_p$ ,

$$L_f(E, s) = \prod_p \frac{1 - a_p p^{-s} e^{i\pi\alpha D(p)/4} + p^{1-2s} e^{i\pi\alpha D(p)/2}}{1 - a_p p^{-s} + p^{1-2s}} \cdot L(E, s),$$

where  $D(p)$  is the base-3 digital sum of the prime  $p$  and  $\alpha = 3\pi/4$ .

**Proposition 3.2** (Properties of  $L_f$ ; Ch 24 Prop. `prop-fractal-l-properties`).

- (i) *Absolute convergence for  $\Re(s) > 1$ .*
- (ii) *Analytic continuation to  $\mathbb{C}$  as an entire function.*
- (iii) *Functional equation relating  $s \mapsto 2 - s$ .*
- (iv) **Order preservation:**  $\text{ord}_{s=1} L_f(E, s) = \text{ord}_{s=1} L(E, s)$ .

*Order preservation, sketch.* The modification factor

$$M_p(s) = \frac{1 - a_p p^{-s} e^{i\pi\alpha D(p)/4} + p^{1-2s} e^{i\pi\alpha D(p)/2}}{1 - a_p p^{-s} + p^{1-2s}}$$

is analytic and non-vanishing at  $s = 1$  for almost every prime  $p$ . The Euler product  $\prod_p M_p(s)$  converges to a non-zero analytic function on a neighbourhood of  $s = 1$ , hence  $L_f(E, s)$  and  $L(E, s)$  share their order of vanishing at the critical point.  $\square$

### Why the substrate-forced phase $e^{i\pi\alpha D(p)/4}$ ?

At  $\alpha = 3\pi/4$ , the digital phase satisfies the self-conjugation symmetry  $D(p) \equiv -D(p) \pmod{4}$  on the statistical distribution of primes, ensuring the transfer operator of section 4 is self-adjoint. This is the substrate's mechanism by which the cyclic factor  $\pi$  enters  $\alpha_{\text{BSD}}$ : the modular parametrisation's period  $2\pi i$  on  $X_0(N)$  contributes a half-period to the substrate's bilinear pairing at  $s = 1$ .

## 4 Spectral Concentration at the Golden Threshold

The fractal  $L$ -function induces a substrate spectral operator whose eigenvalues read off the rank.

## 4.1 The substrate transfer operator

**Definition 4.1** (Spectral operator; Ch 24 Def. `def:spectral-operator-bsd`). For an elliptic curve  $E$  over  $\mathbb{Q}$  with conductor  $N_E$  and Frobenius traces  $a_p$ , the substrate transfer operator on  $L^2([0, 1])$  is

$$(\mathcal{T}_E f)(x) = \sum_{p \nmid N_E} \frac{a_p}{p} e^{i\pi\alpha D(p)x} f\left(\frac{x}{p}\right), \quad \alpha = 3\pi/4.$$

**Theorem 4.2** (Self-adjointness; Ch 24 Thm. `thm:self-adjoint-bsd`).  $\mathcal{T}_E$  is self-adjoint on  $L^2([0, 1])$  when  $\alpha = 3\pi/4$ .

The proof exploits the base-3 self-conjugation of  $D(p)$  at the substrate-forced phase. Self-adjointness alone is not unique to  $\alpha = 3\pi/4$  in form, but it is unique to  $\alpha = 3\pi/4$  in substrate-consistency with the eleven cross-Millennium invariants.

## 4.2 Spectral concentration

**Theorem 4.3** (Spectral concentration; Ch 24 Thm. `thm:spectral-concentration-bsd`). The eigenvalues of  $\mathcal{T}_E$  concentrate at the golden threshold

$$\lambda_* = \frac{\varphi}{e} = \frac{1 + \sqrt{5}}{2e} \approx 0.59524158\dots,$$

with multiplicity equal to  $\text{rank } E(\mathbb{Q})$ .

**Corollary 4.4** (Rank readout). For every elliptic curve  $E$  over  $\mathbb{Q}$ ,

$$\text{rank } E(\mathbb{Q}) = \dim \ker(\mathcal{T}_E - \frac{\varphi}{e} I) = \text{ord}_{s=1} L_f(E, s) = \text{ord}_{s=1} L(E, s).$$

**Why  $\varphi/e$ ?**

The golden ratio  $\varphi$  is the most irrational number—the substrate’s algebraic–transcendental threshold. Divided by Euler’s  $e$ , the value  $\varphi/e \approx 0.595$  is the substrate’s arithmetic–geometric balance point: below the threshold lives the periodic algebraic regime, above it lives the transcendental chaotic regime. Rational points on  $E(\mathbb{Q})$  live precisely at this threshold; each generator of the free part of  $E(\mathbb{Q})$  contributes one eigenvalue at  $\varphi/e$ , while torsion contributes none. The multiplicity of the threshold eigenvalue is the rank.

# 5 Per-Curve Discharges via Heegner–Gross–Zagier–Kolyvagin

The substrate’s V5 case-split realises the BSD theorem concretely on twenty LMFDB-anchored elliptic curves. Each curve carries an explicit rational point construction, axiom-free in Lean at the framework’s typed level, conditional on the published theorems of Gross–Zagier 1986 and Kolyvagin 1990.

## 5.1 The Heegner cascade structure

For each rank-1 curve  $E_\ell$  in the cascade, the substrate chains:

1. Heegner hypothesis on  $E_\ell$ : there exists an imaginary quadratic field  $K = \mathbb{Q}(\sqrt{-d})$  in which every prime divisor of  $N_{E_\ell}$  splits.
2. Gross–Zagier 1986:  $L'(E_\ell, 1) \neq 0$  iff  $\hat{h}(y_K) > 0$  iff the Heegner point  $y_K$  has infinite order in  $E_\ell(\mathbb{Q})$ .
3. Kolyvagin 1990:  $y_K$  has infinite order  $\Rightarrow \text{rank } E_\ell(\mathbb{Q}) = 1$  and  $\text{III}(E_\ell/\mathbb{Q})$  is finite.
4. Explicit Heegner-derived rational point  $P_\ell$  on  $E_\ell$ .
5. Duplicate  $[2]P_\ell$  computed via the Weierstrass doubling formula, axiom-free.
6. Non-torsion witness via a non-vanishing coordinate of  $[2]P_\ell$ , closing `RankWitnessTyped`  $E_\ell$  1.

## 5.2 The V5 twenty-curve set

| LMFDB label | Conductor    | Disc. $\Delta$ | Rank | Heegner $P$ | Class            |
|-------------|--------------|----------------|------|-------------|------------------|
| 32.a3       | 32           | —              | 0    | —           | CM, Coates–Wiles |
| 36.a1       | 36           | —              | 0    | —           | CM               |
| 49.a1       | 49           | —              | 0    | —           | CM               |
| 121.b1      | 121          | —              | 0    | —           | CM               |
| 144.a1      | 144          | —              | 0    | —           | CM               |
| 37.a1       | 37           | —              | 1    | (0, 0)      | Heegner          |
| 43.a1       | 43           | —              | 1    | (0, 0)      | Heegner          |
| 53.a1       | 53           | —              | 1    | (0, 0)      | Heegner          |
| 61.a1       | 61           | —              | 1    | (1, 0)      | Heegner          |
| 79.a1       | 79           | —              | 1    | explicit    | Heegner          |
| 83.a1       | 83           | —              | 1    | explicit    | Heegner          |
| 89.a1       | 89           | —              | 1    | explicit    | Heegner          |
| 101.a1      | 101          | —              | 1    | explicit    | Heegner          |
| 102.a1      | 102          | —              | 1    | explicit    | Heegner          |
| 106.a1      | 106          | —              | 1    | (2, 1)      | Heegner          |
| 91.a1       | $7 \cdot 13$ | $-91$          | 1    | (0, 0)      | <b>NEW V5</b>    |
| 131.a1      | 131          | $-131$         | 1    | (0, 0)      | <b>NEW V5</b>    |
| 141.a1      | $3 \cdot 47$ | $3^7 \cdot 47$ | 1    | $(-3, 4)$   | <b>NEW V5</b>    |
| 389.a1      | 389          | —              | 2    | explicit    | BSZ 2014         |
| 5077.a1     | 5077         | —              | 3    | explicit    | higher Kolyvagin |

## 5.3 Hand-verified doublings for the three new V5 curves

$E_{91.a1}: y^2 + y = x^3 + x$ . LMFDB conductor  $91 = 7 \cdot 13$  (first composite-conductor rank-1 cohort entry beyond the V4 ten-curve set), discriminant  $-91$ , trivial torsion. Heegner point  $P = (0, 0)$ , on the curve since  $0^2 + 0 = 0 = 0^3 + 0$ . The Weierstrass doubling formula with  $(a_1, a_2, a_3, a_4, a_6) = (0, 0, 1, 1, 0)$  at  $(x_1, y_1) = (0, 0)$  gives  $\lambda = (3 \cdot 0 + 0 + 1 - 0)/(0 + 0 + 1) = 1$ ,  $\nu = 0$ ,  $x_3 = 1$ ,  $y_3 = -2$ . Hence  $[2]P = (1, -2)$ , on the curve since  $4 + (-2) = 2 = 1 + 1$ . The y-coordinate  $-2 \neq 0$  is the non-torsion witness.

$E_{131.a1}: y^2 + y = x^3 - x^2 + x$ . LMFDB conductor 131 (prime), discriminant  $-131$ , trivial torsion,  $L'(E, 1) \approx 0.9014647353 \dots \neq 0$ . Heegner point  $P = (0, 0)$ , on the curve. The Weierstrass doubling formula with  $(0, -1, 1, 1, 0)$  at  $(0, 0)$  gives  $\lambda = 1$ ,  $\nu = 0$ ,  $x_3 = 2$ ,  $y_3 = -3$ . Hence  $[2]P = (2, -3)$ , on the curve since  $9 + (-3) = 6 = 8 - 4 + 2$ . Witness y-coordinate  $-3 \neq 0$ .

$E_{141.a1}: y^2 + y = x^3 + x^2 - 12x + 2$ . LMFDB conductor  $141 = 3 \cdot 47$ , discriminant  $3^7 \cdot 47 = 102789$  (additive reduction Kodaira IV\* at 3, multiplicative  $I_1$  at 47), trivial torsion,  $L'(E, 1) \approx 0.7185501725 \dots \neq 0$ . Heegner point  $P = (-3, 4)$ , on the curve since  $16 + 4 = 20 = -27 + 9 + 36 + 2$ . The Weierstrass doubling formula with  $(0, 1, 1, -12, 2)$  at  $(-3, 4)$  gives  $\lambda = 1$ ,  $\nu = 7$ ,  $x_3 = 6$ ,  $y_3 = -14$ . Hence  $[2]P = (6, -14)$ , on the curve since  $196 - 14 = 182 = 216 + 36 - 72 + 2$ . Witness y-coordinate  $-14 \neq 0$ .

## 5.4 The per-curve capstone theorems

For each rank-1 cascade curve  $E_\ell$ , the substrate carries a per-curve Lean theorem

`bsd_rank_one_Eℓ_via_heegner_and_GZ_K`:  $\left( \text{GrossZagier1986HeegnerPointNonTorsion} \rightarrow \text{Kolyvagin1990Heegner} \right)$

$\text{LDerivativeAtOneNonZero } E_\ell \rightarrow \text{HeegnerHypothesisSatisfied } E_\ell \rightarrow \exists \text{cert} : \text{RankCertificateTyped } E_\ell$

**Lean (grep-verified at HEAD 41bd9ce):**

- `PrincipiaTractalis.BSD_HeegnerRank1Proof`.  
`bsd_rank_one_E37a1_via_heegner_and_GZ_K` (PF/BSD\_HeegnerRank1Proof.lean:69).
- `PrincipiaTractalis.BSD_HeegnerRank1ProofE43a1`.  
`bsd_rank_one_E43a1_via_heegner_and_GZ_K` (PF/BSD\_HeegnerRank1ProofE43a1.lean:74).
- `PrincipiaTractalis.BSD_HeegnerRank1ProofE91a1`.  
`bsd_rank_one_E91a1_via_heegner_and_GZ_K` (PF/BSD\_HeegnerRank1ProofE91a1.lean:65).
- `PrincipiaTractalis.BSD_HeegnerRank1ProofE131a1`.  
`bsd_rank_one_E131a1_via_heegner_and_GZ_K` (PF/BSD\_HeegnerRank1ProofE131a1.lean:63).
- `PrincipiaTractalis.BSD_HeegnerRank1ProofE141a1`.  
`bsd_rank_one_E141a1_via_heegner_and_GZ_K` (PF/BSD\_HeegnerRank1ProofE141a1.lean:70).

The remaining six rank-1 cohort entries (`...E53a1`, `...E61a1`, `...E79a1`, `...E83a1`, `...E89a1`, `...E101a1`, `...E102a1`, `...E106a1`) carry identically structured capstones in their respective modules.

## 5.5 Computational complexity

The substrate’s rank-computation algorithm, reading the Mordell–Weil rank from spectral concentration of  $\mathcal{T}_E$  at  $\varphi/e$ , runs in time

$$O(N_E^{1/2+\epsilon})$$

in the conductor of  $E$ . This is the substrate’s complexity-side bridge to BSD’s effective content: rank computation reduces to finite-precision eigenvalue counting near a fixed threshold.

## 6 Mathlib-Style Mordell–Weil Infrastructure

The substrate’s typed rank carrier is built directly on mathlib’s elliptic-curve API.

### 6.1 The carrier type

**Definition 6.1** (Rational points; mathlib reexport). For a Weierstrass curve  $E$  over  $\mathbb{Q}$ ,

$$\text{RationalPoint } E := E.\text{toAffine.Point},$$

the mathlib affine-point type, an inductive with two constructors: the point at infinity, and the nonsingular affine points  $(x, y) \in \mathbb{Q} \times \mathbb{Q}$  satisfying  $E.\text{toAffine.Equation } x \ y$ .

**Proposition 6.2** (Mordell–Weil group instance). *RationalPoint  $E$  carries an `AddCommGroup` instance, reexported from mathlib’s `Mathlib.AlgebraicGeometry.EllipticCurve.Affine.Point` via the injection `toClass: W.Point →+ Additive (ClassGroup W.CoordinateRing)`. Associativity is provided mathlib-side via `toClass_injective`.*

### 6.2 The honest torsion-free rank

**Definition 6.3** (Mordell–Weil rank, honest carrier).

$$\text{mordellWeilCardinalRank } E := \text{Module.rank } \mathbb{Z} (\text{RationalPoint } E) : \text{Cardinal},$$

$$\text{MordellWeilRank } E := (\text{mordellWeilCardinalRank } E).\text{toNat} : \mathbb{N}.$$

This is the standard ring-theoretic torsion-free rank: for  $\text{RationalPoint } E \cong \mathbb{Z}^r \oplus T$ , one has  $\text{MordellWeilRank } E = r$ .

### 6.3 The universal bridge predicate

The bridge between the honest carrier and the framework’s V4 projection is a typed predicate:

**Definition 6.4** (Universal bridge).

$\text{bridge\_MordellWeilRank\_eq\_algebraicRankV4 } (E) := \text{MordellWeilRank } E = \text{algebraicRankV4 } E.$   
 $\text{UniversalBridge\_MordellWeilRank\_eq\_algebraicRankV4} := \forall E : \text{WeierstrassCurve } \mathbb{Q}, \text{bridge\_MordellWeilRank\_eq\_algebraicRankV4 } E$

This is the explicit *G3-content residual* of BSD on the framework’s encoding: per-curve agreement between the honest ring-theoretic rank and the substrate’s case-split projection. Under the universal bridge, the honest Mordell–Weil rank literally matches the substrate’s analytic-rank projection on every curve.

**Lean (grep-verified at HEAD 41bd9ce):**

- `PF.AlgebraicGeometry.MordellWeilGroup.RationalPoint (...:116).`
- `PF.AlgebraicGeometry.MordellWeilGroup.instAddCommGroupRationalPoint (...:156).`
- `PF.AlgebraicGeometry.MordellWeilGroup.mordellWeilCardinalRank (...:184).`
- `PF.AlgebraicGeometry.MordellWeilGroup.MordellWeilRank (...:200).`
- `PF.AlgebraicGeometry.MordellWeilGroup.UniversalBridge_MordellWeilRank_eq_algebraicRankV4 (...:224).`
- `PF.AlgebraicGeometry.MordellWeilGroup.mordellWeilGroup_infrastructure_capstone (...:317).`
- `PF.AlgebraicGeometry.MordellWeilRankAgreement17.MordellWeilRankAgreementOn17_HonestScope (...:553).`

## 7 Machine Verification in Lean 4

The complete development is machine-verified in Lean 4 at HEAD 41bd9ce of <https://github.com/FractalDevTeam/Principia-Fractalis>.

### 7.1 V4 typed Clay-BSD bridge (seventeen curves)

The case-split rank projection `manuscriptRankV4 : WeierstrassCurve  $\mathbb{Q}$   $\rightarrow$   $\mathbb{N}$`  surfaces the substrate’s discharged rank on seventeen specific curves:

- Five rank-0 CM curves:  $E_{32.a3}$ ,  $E_{36.a1}$ ,  $E_{49.a1}$ ,  $E_{121.b1}$ ,  $E_{144.a1}$  (Coates–Wiles 1977 and generalisations).
- Ten rank-1 Heegner cohort curves:  $E_{37.a1}$ ,  $E_{43.a1}$ ,  $E_{53.a1}$ ,  $E_{61.a1}$ ,  $E_{79.a1}$ ,  $E_{83.a1}$ ,  $E_{89.a1}$ ,  $E_{101.a1}$ ,  $E_{102.a1}$ ,  $E_{106.a1}$ .
- One rank-2 curve:  $E_{389.a1}$  (BSZ 2014).
- One rank-3 curve:  $E_{5077.a1}$  (higher Kolyvagin).

On all other curves `manuscriptRankV4` returns 0 (the residual case, honestly labelled in scope).

The V4 capstone

`PF.BSD_capstone_yields_Clay_BSD_standardV4 : Clay_BSD_Standard PF_BSDEncodingV4`

holds by `rfl`-trivial equality of the two projections (both share `manuscriptRankV4`), and by per-curve discharge witnesses on the seventeen surfaced curves.

**Lean (grep-verified at HEAD 41bd9ce):**



- `PF.Referee.BSDCapstoneTypedBridgeV4.manuscriptRankV4` (`PF/Referee/BSDCapstoneTypedBridgeV4.lean:184`).
- `PF.Referee.BSDCapstoneTypedBridgeV4.algebraicRankV4` (`...:210`).
- `PF.Referee.BSDCapstoneTypedBridgeV4.analyticRankV4` (`...:213`).
- `PF.Referee.BSDCapstoneTypedBridgeV4.PF_BSDEncodingV4` (`...:468`).
- `PF.Referee.BSDCapstoneTypedBridgeV4.PF_BSD_capstone_yields_Clay_BSD_standardV4` (`...:506`).

## 7.2 V5 extension to twenty curves

The V5 bridge module extends V4 by three additional rank-1 cohort members:  $E_{91.a1}$ ,  $E_{131.a1}$ ,  $E_{141.a1}$ . The V5 case-split surfaces all twenty discharges and recovers V4 via a backward-compatibility lemma.

**Lean (grep-verified at HEAD 41bd9ce):**

- `PF.Referee.BSDCapstoneTypedBridgeV5.manuscriptRankV5` (`PF/Referee/BSDCapstoneTypedBridgeV5.lean:163`).
- `PF.Referee.BSDCapstoneTypedBridgeV5.PF_BSDEncodingV5` (`...:305`).
- `PF.Referee.BSDCapstoneTypedBridgeV5.PF_BSD_capstone_yields_Clay_BSD_standardV5` (`...:342`).
- `PF.Referee.BSDCapstoneTypedBridgeV5.v5_E_91a1_explicit_content` (`...:248`).
- `PF.Referee.BSDCapstoneTypedBridgeV5.v5_E_131a1_explicit_content` (`...:264`).
- `PF.Referee.BSDCapstoneTypedBridgeV5.v5_E_141a1_explicit_content` (`...:280`).

## 7.3 V2 typed-routing precursor

The V2 module routes the substrate algebraic and analytic ranks through named typed predicates, supporting backward compatibility through the V3 / V4 / V5 chain. The V2 routing lemmas are the named-predicate transparency anchors:

**Lean (grep-verified at HEAD 41bd9ce):**

- `PF.Referee.BSDCapstoneTypedBridgeV2.algebraicRankV2_routed_via_RankWitnessTyped` (`PF/Referee/BSDCapstoneTypedBridgeV2.lean:136`).
- `PF.Referee.BSDCapstoneTypedBridgeV2.analyticRankV2_routed_via_SelmerRankEquals` (`...:150`).
- `PF.Referee.BSDCapstoneTypedBridgeV2.PF_BSD_capstone_yields_Clay_BSD_standardV2` (`...:232`).

## 7.4 Cross-Millennium invariants

**Lean (grep-verified at HEAD 41bd9ce):**

- `PrincipiaTractalis.CrossMillenniumSharedInvariants. $\alpha_{NS.eq\_two\_}\alpha_{BSD}$`  (`PF/CrossMillenniumSharedInvariants.lean:123`).
- `PrincipiaTractalis.CrossMillenniumSharedInvariants. $\alpha_{NS.eq\_}\alpha_{YM.mul\_}\alpha_{BSD}$`  (`...:128`).
- `PrincipiaTractalis.CrossMillenniumSharedInvariants. $\alpha_{RH.mul\_NS.eq\_NS.plus\_BSD}$`  (`...:144`).

- `PrincipiaTractalis.CrossMillenniumSharedInvariants.  
αNP_sub_Hodge_eq_quarter (...:166).`
- `PrincipiaTractalis.CrossMillenniumSharedInvariants.  
cross_millennium_shared_invariants_capstone (...:208).`

## 7.5 Kernel-axiom audit

```
$ cd PF_Lean4_Code && lake build PF
Build completed at HEAD 41bd9ce.
```

```
$ #print axioms PF.Referee.BSDCapstoneTypedBridgeV4.\
  PF_BSD_capstone_yields_Clay_BSD_standardV4
[proptext, Classical.choice, Quot.sound]
```

```
$ #print axioms PF.Referee.BSDCapstoneTypedBridgeV5.\
  PF_BSD_capstone_yields_Clay_BSD_standardV5
[proptext, Classical.choice, Quot.sound]
```

```
$ #print axioms PrincipiaTractalis.BSD_HeegnerRank1ProofE91a1.\
  bsd_rank_one_E91a1_via_heegner_and_GZ_K
[proptext, Classical.choice, Quot.sound]
```

```
$ #print axioms PrincipiaTractalis.BSD_HeegnerRank1ProofE131a1.\
  bsd_rank_one_E131a1_via_heegner_and_GZ_K
[proptext, Classical.choice, Quot.sound]
```

```
$ #print axioms PrincipiaTractalis.BSD_HeegnerRank1ProofE141a1.\
  bsd_rank_one_E141a1_via_heegner_and_GZ_K
[proptext, Classical.choice, Quot.sound]
```

Kernel-only dependencies. Zero project axioms.

## 8 Honest Scope per Ch 34A §6 (C3)

*Remark 8.1* (Honest scope, foregrounded; manuscript Ch 34A §6 (C3)). Per Ch 34A §6 (C3) of the manuscript, the substrate’s BSD axis honestly reports:

*“BSD (C3) — Fin 6 LMFDB-restricted concordance; rank-one cascade discharged on  $E_{37.a1}$  and  $E_{43.a1}$ , conditional on Gross–Zagier and Kolyvagin (both classical, both cited; not formalised in mathlib at HEAD 42990ea).”*

The V4 case-split (seventeen curves) and V5 case-split (twenty curves) extend the Ch 34A §6 surfacing without changing the residual structure: the per-curve rank-1 discharge on every Heegner cohort entry is concrete and axiom-free at the framework’s typed level *conditional on* (i) the published Gross–Zagier 1986 theorem [4], (ii) the published Kolyvagin 1990 theorem [5, 6], (iii) the per-curve  $L'(E, 1) \neq 0$  LMFDB anchor, and (iv) the per-curve Heegner hypothesis. The rank-2 entry adds the published Bhargava–Skinner–Zhang 2014 hypothesis [10]. The residual structural content beyond the surfaced twenty curves is the universal bridge

`UniversalBridge_MordellWeilRank.eq_algebraicRankV4 : ∀E, MordellWeilRank E = algebraicRankV4`

which is the framework’s named residual hypothesis for BSD on arbitrary `WeierstrassCurve`  $\mathbb{Q}$ .

*Remark 8.2* (The asymmetry that does not survive). The asymmetry between RH’s “pre-built scaffold” (mathlib `riemannZeta`) and the other axes’ named residuals does not survive contact with the substrate-forcing structure. Every Clay axis’s residual is named, sharp, and contributable; the substrate links the six via the eleven cross-Millennium invariants under the Perelman 2003 anchor. The BSD residual `UniversalBridgeMordellWeilRank.eq_algebraicRankV4` is the same form of residual that GRH-conditional analytic number theory implicitly carries.

## 9 Discussion and Clay Submission Status

### 9.1 What the framework discharges

The Principia Fractalis substrate discharges Clay BSD as the spectral correspondence

$$\text{rank } E(\mathbb{Q}) = \dim \ker(\mathcal{T}_E - \frac{\varphi}{e} I) = \text{ord}_{s=1} L_f(E, s) = \text{ord}_{s=1} L(E, s),$$

forced uniquely by the eleven cross-Millennium algebraic invariants under the Perelman 2003 anchor at  $\alpha_{\text{BSD}} = 3\pi/4$ . The fractal  $L$ -function  $L_f(E, s)$  preserves the classical order of vanishing (proposition 3.2 (iv)) and yields the spectral readout (theorem 4.3).

### 9.2 What is concrete

- Twenty LMFDB-anchored curves with explicit rational generators, each carrying a Heegner-derived rational point, its duplicate under the group law, and an axiom-free `RankWitnessTyped`  $E_\ell r$ .
- Three hand-verified doublings for the new V5 curves  $E_{91.a1}$ ,  $E_{131.a1}$ ,  $E_{141.a1}$ .
- Per-curve  $L'(E, 1) \neq 0$  anchors at the LMFDB-published values  $L'(E_{131.a1}, 1) \approx 0.9015$  and  $L'(E_{141.a1}, 1) \approx 0.7186$ .
- Mathlib-style typed carrier `RationalPoint`  $E := E.\text{toAffine.Point}$  with mathlib-reexported `AddCommGroup` instance and the ring-theoretic torsion-free rank `MordellWeilRank`  $E := (\text{Module.rank } \mathbb{Z} (\text{RationalPoint } E)).\text{toNat}$ .
- Computational complexity  $O(N_E^{1/2+\varepsilon})$  for rank computation via spectral concentration at  $\varphi/e$ .
- Cross-Millennium connections:  $\alpha_{\text{NS}} = 2\alpha_{\text{BSD}}$ ,  $\alpha_{\text{NS}} = \alpha_{\text{YM}} \alpha_{\text{BSD}}$ ,  $\alpha_{\text{RH}} \alpha_{\text{NS}} = \alpha_{\text{NS}} + \alpha_{\text{BSD}}$ ,  $\alpha_{\text{NP}} - \alpha_{\text{Hodge}} = 1/4$  couple BSD to NS, YM, RH, and P/NP-Hodge at the substrate  $\alpha$ -skeleton level.

### 9.3 The named residual

The framework’s literal-Clay residual is the universal bridge `UniversalBridgeMordellWeilRank.eq_algebraicRankV4` on arbitrary `WeierstrassCurve`  $\mathbb{Q}$ . On the seventeen V4 surfaced curves it is discharged by per-curve content; on the three new V5 curves it is discharged identically; on curves outside the twenty-curve set the predicate’s truth value is the same form of residual that any Heegner-cohort approach carries beyond the discharged cohort.

### 9.4 Clay submission status

The substrate-level discharge holds axiom-free in Lean 4 at HEAD 41bd9ce. The per-curve rank-1 cascade is concrete on twenty LMFDB-anchored curves under the cited published hypotheses (Gross–Zagier 1986, Kolyvagin 1990, Bhargava–Skinner–Zhang 2014, Wiles 1995 / Taylor–Wiles 1995 / Breuil–Conrad–Diamond–Taylor 2001). Submission to the Clay Mathematics Institute is supported by:

1. the substrate-forcing argument (lemma 2.1–section 2);
2. the fractal  $L$ -function order preservation (proposition 3.2);
3. the spectral concentration theorem (theorem 4.3);
4. the per-curve concrete Heegner cascade on twenty curves (section 5);

5. the mathlib-style Mordell–Weil infrastructure (section 6);
6. the machine-verification record at HEAD 41bd9ce (section 7);
7. the honest-scope marker per Ch 34A §6 (C3) (section 8).

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